

Session 1: Scanning and Filtering a Source Program

I. OBJECTIVES

To develop a program which can filter comments and white space characters from a source program

II. DEMONSTRATION OF USEFUL RESOURCES

Extracting the sequence of occurrences of a specified character from a source program

Sample Input: datafile1.c

```
datafile1.c
#include <stdio.h>
int main(void)
{
    FILE *p1,*p2; int c;

    p1 = fopen("datafile1.c", "r");
    p2 = fopen("parentheses.txt","w");

    if(!p1) printf("\nFile can't be opened!");
    else {
        while((c = fgetc(p1)) != EOF) {
            if ((c == '(') || (c == ')'))
                fputc(c, p2); } }
    fclose(p1);
    fclose(p2);

    p2 = fopen("parentheses.txt","r");
    while((c=fgetc(p2))!=EOF)
        printf("%c",c);
    fclose(p2);

    return 0;
}
```

Output of the program: 00000((0))((00))0000((0))00

III. LAB EXERCISE

1. Write a program to print the header files used in a source program.

Sample Input: *input.c*

```
#include <stdio.h>
int main()
{
    // printf() displays the string inside quotation
    printf("Hello, World!");
    return 0;
}
```

Sample Output: *stdio.h*

2. Write a program to add line numbers to a source program.

Sample Input: *input.c*

Sample Output:

```
1: #include <stdio.h>
2: int main()
3: {
4: // printf() displays the string inside quotation
5: printf("Hello, World!");
6: return 0;
7: }
```